# OVERVIEW

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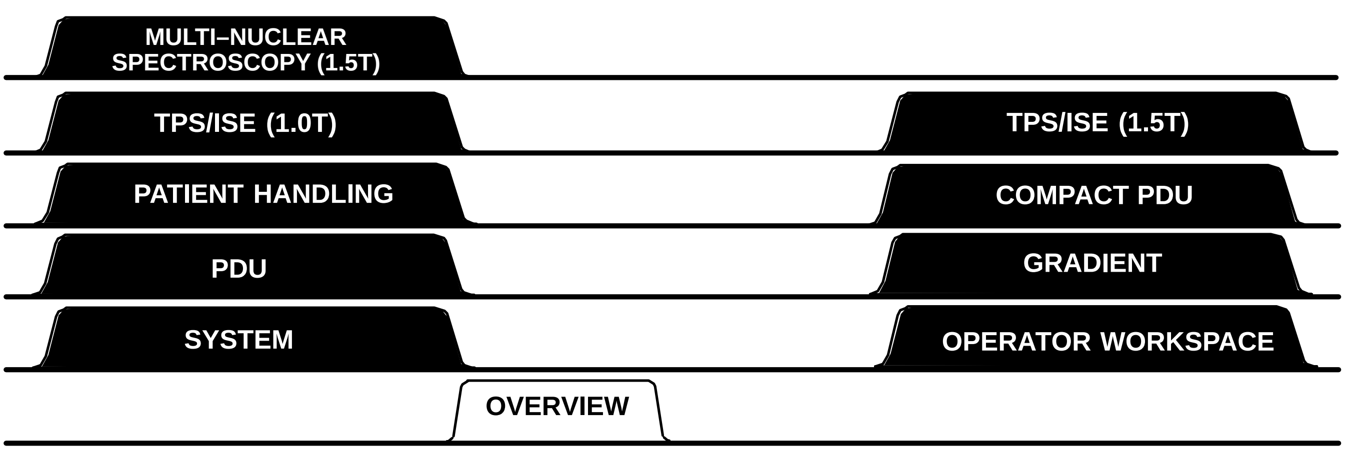
# SECTION 1 – PURPOSE

**Note**
This revision of Direction 2153389 is compatible with Release 8.X of Signa® Horizon™ hardware.

The purpose of this document is to provide an overview of the hardware which makes up the MR Signa System, and a quick means to troubleshoot major signals within the system.

It is recommended that you page through this manual to become familiar with its content and organization, beginning with System tab Section 1, OVERALL SYSTEM. This section shows how the major subsystems are grouped by cabinet, and the main communication lines between them. System tab Section 2, POWER AND ON/OFF CONTROL, contains a block diagram of the main power distribution and the different power “ON” modes.

The remaining block diagrams are organized into sections which correspond to major subsystems. Some diagrams show hardware from another subsystem in order to complete an entire signal flow route. For such exceptions, the board/module title of the circuitry shown will generally help identify the applicable subsystem. See Illustration 1–1 for tab organization of this manual.




**TAB ORGANIZATION FOR DIRECTION 2153389**
ILLUSTRATION 1

**SECTION 1 – PURPOSE (continued)**

These block diagrams, along with provided diagnostics, are intended to be the primary tools for troubleshooting the MR System from a system or subsystem level on down to the specific module or board level. Since some of the RF subsystem block diagrams are presently insufficient for troubleshooting to the FRU (Field Replaceable Unit) level, schematics have been provided to supplement those block diagrams. When using the diagnostics, the block diagrams can provide a better understanding of what circuitry the diagnostics are testing, and how to further isolate a problem if required.

Table 1–1 identifies which tabs correspond to 1.5T, and 1.0T systems.

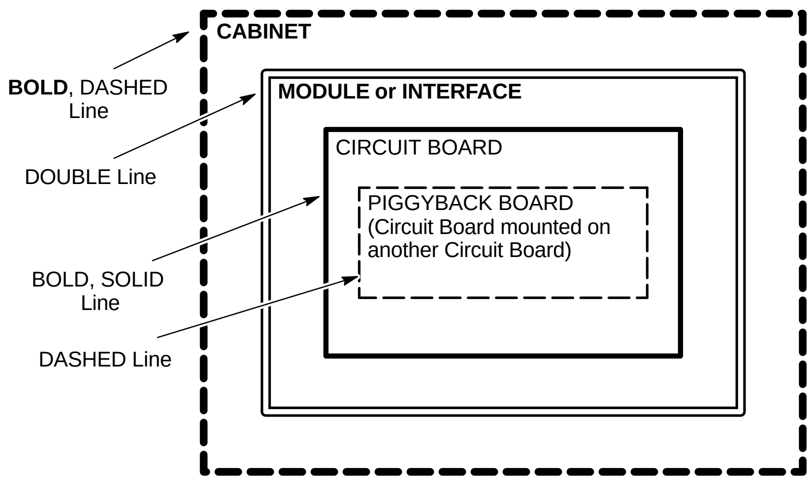
TABLE 1–1
**DIRECTION 2153389 TAB COMPATIBILITY LIST**

| TAB | SIGNA (RELEASE 8.X) 1.5T COMPATIBLE | SIGNA (RELEASE 8.X) 1.0T COMPATIBLE |
|---|---|---|
| OVERVIEW | Yes | Yes |
| SYSTEM | Yes | Yes |
| OPERATOR WORKSPACE | Yes | Yes |
| GRADIENT | Yes | Yes |
| PATIENT HANDLING | Yes | Yes |
| RF COILS | Yes | Yes |
| TPS/ISE (1.5T) | Yes | For some schematics |
| TPS/ISE (1.0T) | No | Yes |
| MULTI–NUCLEAR SPECTROSCOPY (1.5T) | Yes | No |

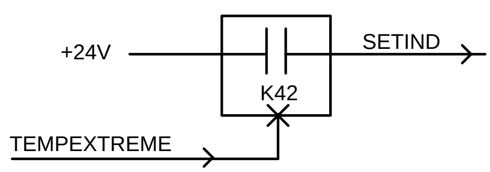
# SECTION 2 – STANDARD CONVENTIONS

The following standard conventions are used within the block diagrams:

- The type of line used to enclose a major box on a block diagram indicates the following:




- The component designator for cabinets, modules, circuit boards, etc. is enclosed in "( )" after or below their name on the block diagram.
- Logic levels: A "0" indicates a logic "low" level (typ. <0.8VDC), a "1" indicates a logic "high" level (typ. > 2.0VDC).
- An "*" is used to identify an "active low" signal (e.g. "INT*" signal when at a logic "0" indicates that an interrupt is present; a logic "1" level indicates that no interrupt is present).
- A "#/" indicates the number of wires in a cable (e.g. —9/— indactes 9 wires).
- Signal flow is generally from left to right on a block diagram. Arrows are used to identify exceptions to this rule and to aid in identifying input/output signal lines on the diagram.
- The "x" symbol (e.g. ——x) shown on boxes in a block diagram are used to indicate that the function within the box is enabled by the "active" signal on this input line.




For this example, when signal TEMPEXTREME goes active ("1"), relay K42 will close (become enabled) and the signal SETIND now becomes active to set a temperature indicator.

# SECTION 3 – ABBREVIATIONS

**Note**

This list has been compiled from a wide variety of sources. Not all of these abbreviations will apply to your system. For an electronic copy of over 1000 acronyms, see *Direction 2124201–2.*

## A

AC = Alternating Current
ACK = Acknowledge
ACQ = Acquisition
A/D = Analog–to–Digital
ADC = Analog to Digital Converter
ADD or ADDR = Address
ADJ = Adjust
ADV = Advance
AGC = Automatic Gain Control
AI = Applications Interface, Artificial Intelligence
ALGN = Align
ALU = Arithmetic Logic Unit
AM = Amplitude Modulation
AMP = ampere, amplifier
A/N = Alpha–numeric
ANSI = American National Standards Institute
AP = Array Processor
APM = Analog Power Monitor
APS = Auto Pre–Scan
APSGT = Auto Pre–Scan Gain Control
AQT = Auto QTUNE
ARCH = Archive/Remove Process
ASC = Automated Service Center
ASCII = American Standard Code for Information Interchange
ASM or ASSY = Assembly or Analog Service Module
ASM = Analog Service Module
ATTEN = Attenuator
AUTO = Automatic
AUX = Auxiliary

## B

B = magnetic field
BATT = Battery
BB = Buffered Data or Broadband
BBT = BroadBand Tool
BC = Body Coil
BD = Board
BDS = Buffered Data Strobe
BLK = Black
BNC = BNC Connector
BRT = Brightness
BUF = Buffer
B/W = BandWidth

## C

C = centigrade, Celsius
CAB = Cabinet
CAL = Calibration
CALC = Calculate
CATTEN = Course Attenuation
CB = Circuit Breaker
CCW = Counter Clockwise
CD–ROM = Compact Disk Read–Only Memory
CE = Chip Enable
CERD = Combined Exciter, Receiver, DAB
CF = Center Frequency
CFA = Center Frequency Adjust
CHAN or CHNL = Channel
CHK = Check
CKT = Circuit
CLR = Clear
CLI = Command Line Interpreter
CMD = Command
CMMR = Common Mode Rejection Ratio
CMOS = Complimentary Symmetry Metal Oxide Semiconductor
CMP = Compare or Compressor
CM/PM = Communications Manager / Power Monitor Board
CNT = Contrast
COAX = Coaxial
COF = Cut Off Frequency
CON or CONS = Console
CONFIG = Configuration
CONT = Continuous

CONV = Converter
CPD = Communication Pin Driver
CPU = Central Processing Unit
CRT = Cathode–Ray Tube
CT = Computed Tomography
CTL = Clock Time Control
CTRL = Control
CV = Configuration Variable
CW = Clockwise
CYL = Cylinder

## D

D/A = Digital–to–Analog
DAC = Digital to Analog Converter
dB = Decibel
dBm = Decibels above or below one Milliwatt
DC = Direct Current
DCI = Digital Control Interface
DCKSTP = Dock Stop
DD = Dynamic Disable
DDS = Direct Digital Synthesizer
DDSB = Dynamic Disable Switch Box
DEC = Decimal
DECC = Digital Eddy Current Compensation
DEG = Degree(s)
DEMUX = Demultiplexer
DEV = device
DIA = Diagnostic Imaging Accessories
DIAG = Diagnostic
DIFF = Differential
DIG = Digit
Dig B0 = Digital B0 Adjustment
DIP = Dual In–line Package
DIR = Direction
DIS = Disable
DMA = Direct Memory Access
DMAC = Direct Memory Access Controller
DMM = Digital Multi–Meter
DPR or DPRAM = Dual Port Random Access Memory
DQA = Daily Quality Assurance
DRAM = Dynamic Random Access Memory
DRC = Dynamic Ram Controller
DRVR = Driver
DSP = Digital Signa Processor
DTS = Display touch Screen
DVM = Digital Volt Meter
DWG = Drawing

## E

ECC = Extended Cursor Control or Error Checking and Correction
ECG = Electro–Cardiogram
EDM = Equipment Diagnostic Monitor
EEPROM = Electrically Erasable Programmable Read Only Memory
EFB = Envelop Feedback
EPI = Echo Planer Imaging
e.g. = for example
EM or EMERG = Emergency
EMI = Electro–Magnetic Interference
EN or ENA = Enable
ENV = Envelope
EOS = End Of Sequence
EOT = End Of Travel
EOW = End Of Waveform
EPO = Emergency Power Off
EPROM = Erasable Programmable Read Only Memory
ERU = Emergency Rundown Unit
EXER = Exerciser

## F

f = farad
F = Fahrenheit, Fuse (designator)
FBD = Functional Block Diagram
FBK = Feedback
FC = footcandle
Fc = Center Frequency
FDBK = Feedback
FDO = Future Delivery Order
FE = Field Engineer
FET = Field–Effect Transistor
FF or FIFO = First In – First Out
FFD = Full Field Distortion
FFT = Fast Fourier Transform
FID = Free Induction Decay
FIDAT = FIFO In Data
FIFO or FF = First In – First Out
FILT = Filter
FM = Frequency Modulated
FMI = Field Modification Instruction
FOV = Field Of View
FPU = Floating Point Unit
FREQ = Frequency

FRESBECC = Frequency Shift B0 Eddy Current Compensation
FRNT = Front
FRU = Field Replaceable Unit
ft = foot
FT = Feet
FTP = File Transfer Protocol

## G

G = gauss
G/A = Gradient Amplifier
GAP = Gradient Amplifier Processor
GASM = Gram Analog Service Module
G(ASM) = GASM and ASM
GEN = Generator
GEPI = Gradient Echo Planer Imaging
GIF = Gradient Interface
GIP = Gradient Interface Processor
GND = Ground
GPIB = General Purpose Interface Board
GPM = Gradient Power Module (aka 8651)
GRAD = Gradient
GRAM = Gradient Ramp Accelerator Module
GRN = Green
GRY = Grey
GP = Gradient Processor

## H

H = henry
H = Hexadecimal
HC = Head Coil
HD = Head
He = Helium
He–ALARM = Helium Alarm
HEX = Hexadecimal
He–WARNING = Helium Warning
HLFC = Host Load From Cold
HORIZ or HORZ = Horizontal
HPC = Hardware and Protocol Control
HPFS = Host Platform Software
hr = hour
HSN = High Speed Network
HSS (D) = High Speed Serial Data Link
HTR = Heater
HV = High Voltage
HV PSU = High Voltage Power Supply Unit
HVAC = Heating, Ventilation, & Air Conditioning
Hz = hertz

## I

I = Current or Phase or Real
IAMP = Amplifier Current (Gradient)
IC = Independent Console, Integrated Circuit
ICD = Installation Certification Document
ICL = Coil Overcurrent (Gradient)
ID = inner diameter
ID = Identity, Identifier
i.e. = that is to say
IEC = International Electrical Code
IEEE = Institute of Electrical & Electronics Engineers
I/F = Interface
IGRAD = GRAM HSSD D/A converter voltage output
in. = inch
INFO = Information
INS, INST, or INSTR = Instruction
INT or INTR = Interrupt
I/O = Input/Output
IP = Image Processor or Plate Current (RF amp) or Instruction Processor or Internet Protocol
IPA = Intermediate Power Amplifier
IPC = Interprocess Communication
IPG = Integrated Pulse Generator
IQ = Image Quality
IRQ = Interrupt Request
ISA = Interrupt Start Address
ISE = Integrated Systems Electronics
ISI = Inter–Sequence Interrupt
ISO = Isolator or Isolation
IVI = Interactive Vascular Imaging
IVME = Internal VME

## J

J = Jack Connector (designator)
JP = Jumper

### K

K = Kelvin (unit of temperature) or Kilo (Thousand)
KB = one thousand bytes of memory
KCal = kiloCalorie
kg = kilogram
kHz = kilohertz
KVA = kilo Volt Amperes
kW = Kilowatt

### L

L or LFT = Left
LAN = Local Area Network
lb = pound
LBK = Loopback
LCA = Logic Cell Arrays
LED = Light Emitting Diode
LFC = Load From Cold
LHe =Liquid Helium
LIN = Linear
LM or LMK = Landmark
LMP = Lamp
LNK = Link
LO = Local Oscillator
LOEP = List of Effective Pages
LONG = Longitudinal
LVLE = Low Voltage, Low Energy

### M

m = meter
M = mega – one million
mA = millamps
MAG = Magnet
MB = megabyte
MCD = Multi–Coil switch Driver
MDS = Multidrop Serial interface
MEM = Memory
Mfg. = Manufacturing
MFU = Multi–Format Unit
MGMNT = Management
mGRAM = mini Gradient Ramp Accelerator Module
MHz = megahertz
MIF = Master Interface
mm = millimeter
MOD = Modifier
MON = Monitor
MOSFET = Metal Oxide Semiconductor Field Effect Transistor
MOV = Metal Oxide Varistor
MPS = Manual Pre–Scan
MR = Magnetic Resonance
MSG = Message
MTR = Motor
MUX = Multiplexer
mV = millivolts
MW = megawatt
mW = milliwatt

### N

N = Negative
N2 = Nitrogen
n.c. = normally closed
NEC = National Electrical Code
Ni–Cad = Nickel Cadmium
n.o. = normally open
NO. = Number
NOPROC = No Processing
NOREC = No Reconstruction
nS = Nanosecond

### O

O2 = oxygen
OC = Operator’s Console
OD = outside diameter
OPI = Oblique Plane Imaging
OPTO = Optical
ORG = Orange
OT = Overtemp
OV = Over voltage
OVR = Over
OVRLD or OL = Overload
oz = ounce
OW = Operator Workspace

### P

P = Positive
PA = Patient or Power Amplifier
PAC = Physiological Acquisition Controller
PAL = Programmable Array Logic or Patient Alignment Lights
PAN = Primary Archive Node
PAT = Patient
PCI = Programmable Communications Interface
PCT = Pulse Controller Task
PDI = Product Delivery Instruction
PDU = Power Distribution Unit
PEN = Penetration
PIN = Positive–Intrinsic–Negative
PIO = Programmed Input/Output
PIXEL = Picture Element
PM = Phase Modulation, Periodic Maintenance
P/N = Positive/Negative
PNL = Panel
POS = Position
POT = Potentiometer
P–P = Peak to Peak
ppm = parts per million
PREAMP = Preamplifier
PROC = Process, Processor
PROG = Program, –ed, –able
PROM = Programmable Read Only Memory or Program
PS = Power Supply
PSD = Pulse Sequence Database
PSG = Pulse Sequence Generator
PSU = Power Supply Unit
PSS = Power Supply System Board
PUR = Purple
PVC = Polyvinylchloride
PW = Pulse Width
PWA = Printed Wiring Assembly
PWB = printed wiring board
PWR = Power

### Q

Q = Quadrature, Quality Factor or Imaginary
Qty = Quantity
QUAD = Quadrature or Quadrant

### R

R or RT = Right
RAM = Random Access Memory
RCV = Receive
RD = Read
RDY = Ready
RECON = Reconstruction
REF = Reference or Reflected
REG = Register or Regulator
RET or RTN = Return
REV = Revision or Reverse
RF = Radio Frequency
RFAMP = Radio Frequency Amplifier
RFI = Radio Frequency Interference
RF/PEN = Combined RF/Penetration Cabinet
RFSC = RF System Controller
RGB = Red Green Blue
RLA or RLY = Relay
rms = root mean square
ROI = Region of Interest
ROM = Read Only Memory
RSR = Register
RST = Reset
R/W = Read/Write
RX = Receive
RXD = Receive Data

### S

S = Switch (designator)
SACK = Slave Acknowledge
SAG = Sagittal
SAR = Specific Absorption Rate
SARR = Stand–Alone Retrospective Reconstruction
S/C = Superconducting
SCA = Scan Control Assembly
SCB = Signal Collection Board
SCR = Silicon Controlled Rectifier
SCSI = Small Computer System Interface
SEC = second
SEG = Segment
SEL = Select
SEPI = Spin Echo Planer Imaging
SGI = Silicon Graphics Inc.
SIG = Signal
SII = ”S–2” (type of magnet)
SIII = ”S–3” (type of magnet)
SIMMS = Single In–line Memory Module
SIP = Single In–line Package

SL = Slice
SLE = System Level Exerciser
SMPTE = Society of Motion Picture and Television Engineers
SN = Service Note
SNR = Signal–to–Noise Ratio
SPI = Serial Peripheral Interface
SPT = System Performance Test
SPU = Signal Processing Unit
SRAM = Static Random Access Memory
SRF = Solid State RF Amplifier SGA
SRI = Scan Room Interface
SRV = Service
SSM = System Support Module
SSP = Sequence Synchronous Protocol
SST = Small Sample Test
STAB = Stability
STAT = Status
STD = Standard
STD DEV = Standard Deviation
SUB–D = Subminiature D (connector)
SW = Switch or Software
S/W = Software
SYNC = Synchronous
SYS = System
SYSID = System Identification

## T

T = tesla
TBD = To Be Determined
TBL = Table
TEMP = Temperature
TFU = Thyristor Firing Unit
TG = Transmit Gain
THD = TPS Host Diagnostic
TNF = Transient Noise Filter
TP = Test Point (designator)
TPS = Transceiver Processing and Storage
TR or T/R = Transmit–Receive
TRMAP = Transmit/Receive Map
TS = Terminal Strip
TST = Test
TTL = Transistor to Transistor Logic
TX = Transmit
TXD = Transmit Data

## U

UART = Universal Asynchronous Receiver–Transmitter
uf = microfarad
UFI = Ultra Fast Imaging

UNBLK = Unblank
uP = Microprocessor
UPLMT = Up Limit
USART = Universal Synchronous/Asynchronous Receiver–Transmitter
uS or usec = microsecond
UV = Under voltage

## V

VAC = Volts Alternating Current
VAR = Variable
VBUS = Bus Voltage
VCR = Video Cassette Recorder
VCNTRL = Control Voltage for 8645 Techrons
VDC = Volts Direct Current
VERT = Vertical
VFILT = GRAM Output Voltage
VIN = DAC ldi/dt to command the GRAM PWM voltage
VME = Bus Standard Veras Module Europe
VMIF = MIF Control Voltage
Vout–H/L = Power Module High/Low Voltage Output
Vp–p = Volts Peak–to–Peak
VRMS = Volts, Root Mean Square
VSA = VSB Address
VSB = VME Subsystem Bus
VTAC = Vacuum Tube Amplifier Cavity
VTR = Video Tape Recorder

## W

WARP = Waveform And Rotation Processor
WR = Write
WHT = White
WND = Window
WORM = Write Once Read Many
WRT = Write

## X

XCVR = Transceiver
XFMR = Transformer
XMT or XMIT = Transmit

## Y

YEL = Yellow
YMS = Yokogawa Medical Systems
YP = Yellow Pages (Sun Computer)